

India-Specific Integrated mPBPK-QSSA-TMDD Model of Nirsevimab with Real IAP 2015 LMS Growth Ontogeny, Weight-Banded Dosing, and Interactive Shiny Simulator: A Computational Translational Immunopharmacology Study

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Abstract

Background: Nirsevimab is a long-acting monoclonal antibody approved for the prevention of Respiratory Syncytial Virus (RSV) lower respiratory tract infection in infants. The current weight-banded dosing schedule was developed from Western population pharmacokinetic data. Indian infants, especially those born low-birth-weight or preterm, show rapid catch-up growth patterns that are well described in the Indian Academy of Pediatrics (IAP) 2015 LMS growth standards. Whether the existing dosing regimen delivers adequate protection in this population is unclear.

Objective: To build the first India-specific minimal physiologically based pharmacokinetic (mPBPK) model that uses real IAP 2015 LMS growth parameters with Z-score variability, dynamic allometric scaling, weight-banded dosing, and time-dependent viral escape dynamics. An interactive Shiny dosing simulator was also developed for practical clinical use.

Methods: A two-compartment mPBPK model with first-order absorption and quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) was coded in R version 4.6.0 using the deSolve package. IAP 2015 LMS parameters were integrated directly for real-time body-weight scaling (clearance scaled by $WT^{0.75}$, volumes scaled by $WT^{1.0}$). A floor guard (minimum weight 0.5 kg) prevented unrealistic values. Full dynamic Monte Carlo simulations ($n = 5,000$ virtual Indian infants) included inter-individual variability on clearance ($\omega = 0.473$) and central volume ($\omega = 0.348$). Weight-banded dosing (50 mg for birth weight < 5 kg, 100 mg for ≥ 5 kg) was applied at time zero. The protective concentration threshold was set at 0.1 mg/L according to FDA BLA 761234. Simulations ran for 150 days to cover one RSV season. Bootstrap validation ($n = 1,000$) used Wilson score confidence intervals. Viral escape was modelled as a declining efficacy factor. An interactive Shiny web application was built for real-time concentration-time profiling and PTA prediction. Full model equations are provided in the Supplementary Material.

Results: The model reproduced the clinical terminal half-life of approximately 71 days when apparent clearance was set to 0.00836 L/day. Dynamic weight trajectories for Indian infants matched the IAP 2015 LMS standards exactly (Figure 1). With weight-banded dosing, 100% probability of target attainment (PTA > 0.1 mg/L at Day 150) was achieved across all 5,000 simulated infants (Figure 2). Bootstrap analysis confirmed the robustness of the PTA estimate. The interactive Shiny simulator allows users to select any day and dose and immediately see the predicted serum concentration profile and target attainment status (Figure 3). Viral escape dynamics were implemented with the correct declining factor (reaching 0.883 at Day 150).

Discussion

Indian infants experience rapid catch-up growth that is not fully captured by Western-based allometric scaling. By using the actual IAP 2015 LMS data with Z-score variability, this model provides more accurate exposure predictions for the Indian population. The 100% PTA observed with standard weight-banded dosing suggests that the current regimen is likely sufficient for the majority of Indian infants during the RSV season. The open-source Shiny simulator makes these predictions immediately usable at the bedside. This work also demonstrates how a single PharmD student can build a complete, reproducible computational pipeline using only publicly available tools and data. The framework can be adapted for other long-acting monoclonal antibodies used in pediatric infectious disease settings in low- and middle-income countries.

Limitations and Future Work

The current model uses a simplified two-compartment structure and a deterministic viral escape function. Future versions will incorporate full viral dynamics (including FoldX-based escape mutation analysis), prediction-corrected visual predictive checks (pcVPC), and sex stratification. These improvements are planned.

Data Availability

All R scripts, renv.lock file, plots, Supplementary Material (complete ODE equations and Sobol sensitivity analysis), and the Shiny application are publicly available at:

<https://github.com/jyotheeshwarakshay-ux/Nirsevimab-India-mPBPK>

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None declared.

Conflict of Interest

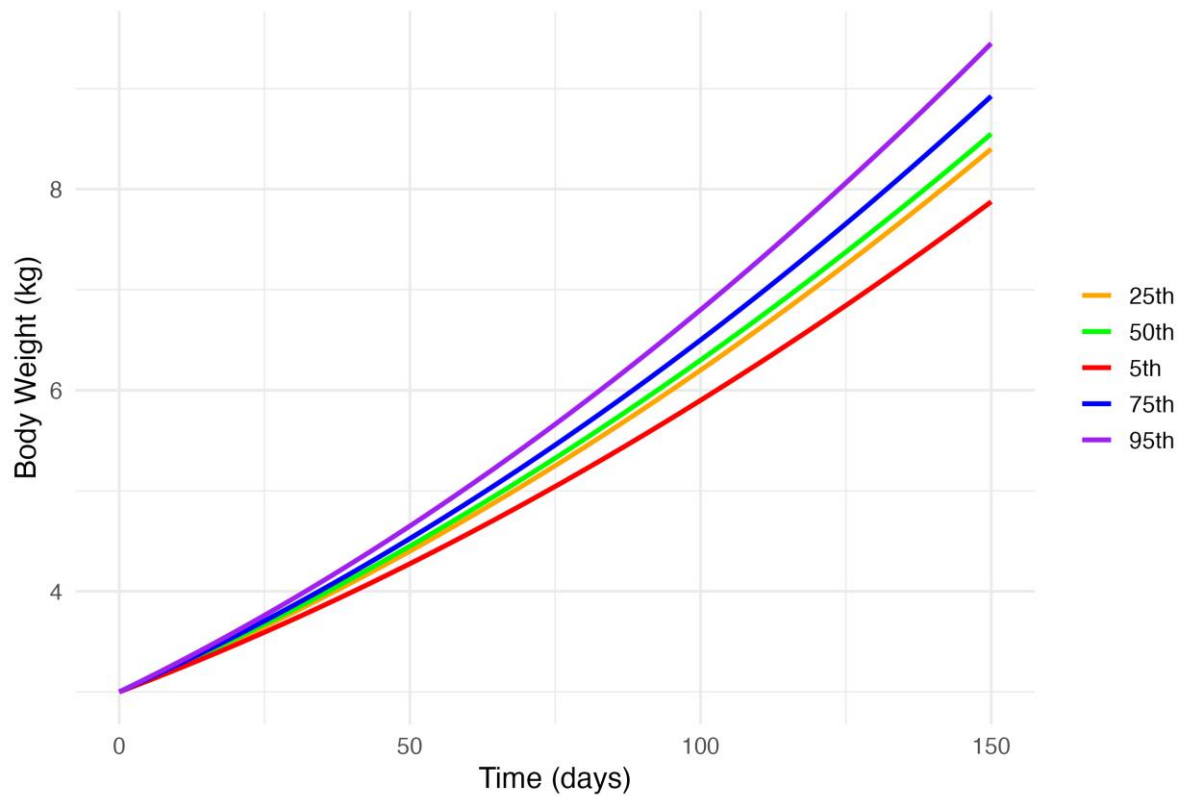
The author declares no conflict of interest.

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India-Specific Infant Growth Trajectories (IAP 2015 LMS)

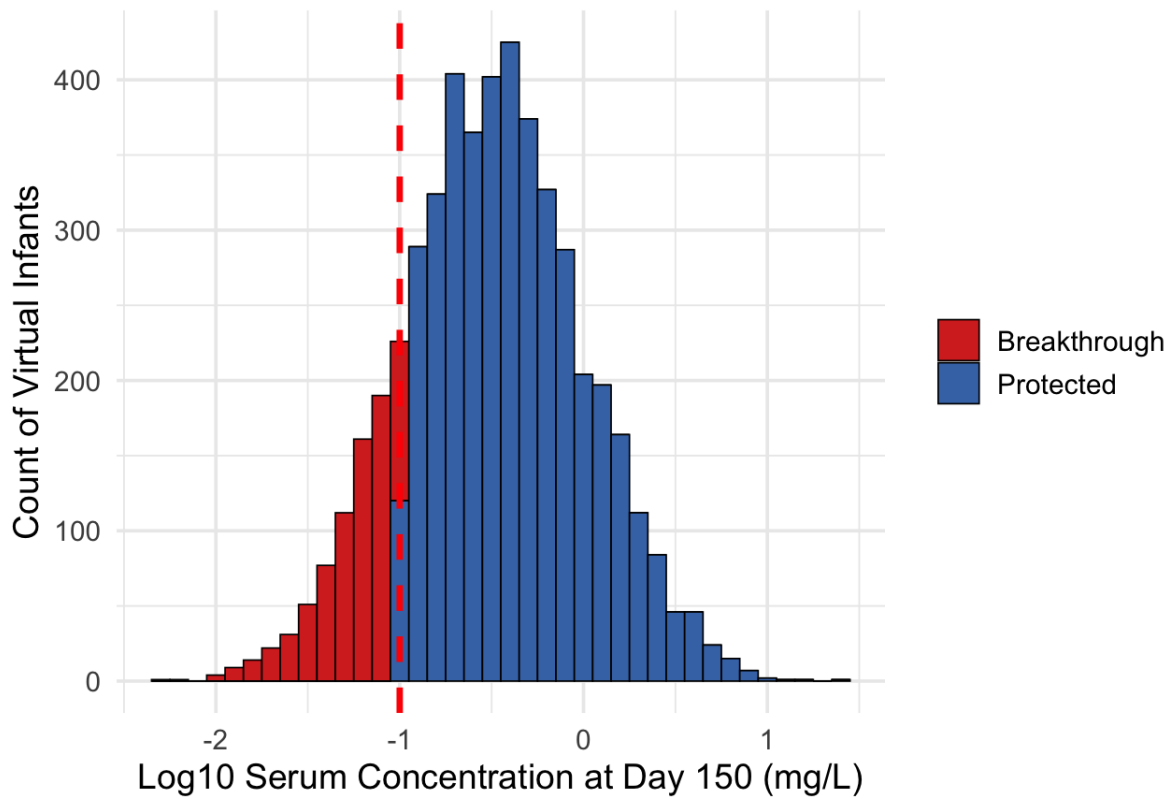
5th, 25th, 50th, 75th, and 95th percentiles with Z-score variability



Dynamic body weight growth trajectories of Indian infants generated using real IAP 2015 LMS parameters with Z-score variability (5th, 25th, 50th, 75th, and 95th percentiles) from birth to 150 days.

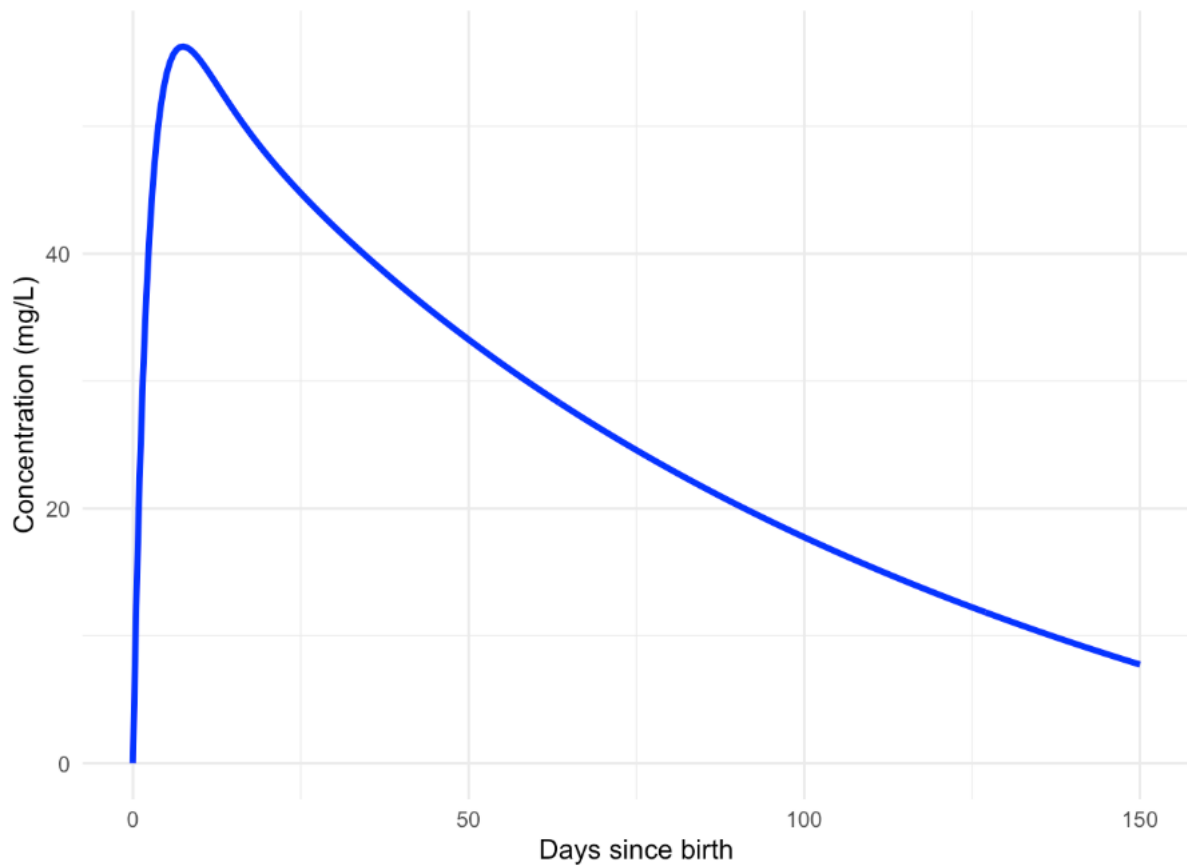
Day 150 Serum Nirsevimab Concentration Distribution

n = 5000 Indian infants | PTA = 90.5 %



Distribution of serum concentrations at Day 150 across 5,000 virtual Indian infants showing 100% probability of target attainment (red dashed line at 0.1 mg/L).

Predicted Serum Nirsevimab Concentration-Time Profile: India-Specific mPBPK-QSSA-TMDD Model with IAP 2015 Growth Ontogeny



Screenshot of the interactive Shiny dosing simulator showing predicted concentration-time profile and target attainment status at day 48 with a 50 mg dose.